

LINEAR MODEL SELECTION – TOWARDS A FRAMEWORK USING A MIXED INTEGER LINEAR PROGRAMMING APPROACH

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Summary: This paper aims at making a contribution to the development of robust linear models by examining the feasibility of data elimination and discarding of explanatory variables simultaneously through the use of mathematical programming techniques. The prediction accuracy of models obtained by such a method is considered via a bootstrap approach and examples are provided to illustrate the characteristics of the proposed procedure.

1. Introduction

Managerial decisions are based, to a large extent, on the evaluation and interpretation of data. One of the most popular ways to investigate possible relationships in a given data set is the use of a linear regression model. Fitting linear regression functions to data assists managers (and researchers) to solve problems by exploring for patterns, relationships and often errors in data sets. These patterns and relationships can then be used to guide the process of decision-making and ideally for forecasting the effect of those decisions.

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Consider the standard linear model

$$y = X\beta + \varepsilon$$

where y is an $n \times 1$ vector of observed values, X is an $n \times s$ given matrix of values where each column vector corresponds to a predictor, β is an $s \times 1$ vector of unknown parameters and ε is an $n \times 1$ vector of (random) errors, ε_i . It is assumed that the ε_i 's are independently distributed continuous random variables with $E(\varepsilon_i) = 0$ and $Var(\varepsilon_i) = \sigma^2 > 0$. β is usually estimated by employing the least squares error criterion.

This model is frequently used to predict values for y . The accuracy of these predictions is then qualified by finding an estimate of the error. Hypothesis testing is usually done to determine the suitability of the overall model as well as the significance of the regression coefficients $\hat{\beta}_i$ obtained. Researchers often try to simplify the model by eliminating predictors using these tests. Sometimes a more pragmatic view is taken by selecting a subset of variables that gives a reasonable fit of the regression equation – a method typically based on measures of fit such as R^2 (adjusted). In addition, researchers have also found it desirable to screen data carefully for errors that can influence the validity of their fitted equations or to highlight cases that need to be investigated (inconsistent with the model fitted to the rest of the cases).

The linear model has been well researched by a large number of researchers over the years. One of the challenges of this model that is still being researched, is the decision to choose elements from a pool of potential predictors that are most suitable for inclusion and to identify data points that should be treated as uncertain or faulty. Many techniques have been proposed to achieve this. Amongst others, Beale, Kendall and Mann (1967), Cook (1977), Belsley, Kuh

and Welsch (1980), Hawkins (1980), Rousseeuw and Leroy (1987), Leger and Altman (1993), Murtaugh (1998) and Choongrak and Soonyoung (2000) contributed to the development of measures and techniques to address the challenge. Most of the approaches used are based on a sequential process whereby predictors are first selected and then uncertain data (outliers) are detected (also a sequential process) or vice versa. This often leads to models that depend upon the order in which these activities were performed. There is a lot less discussion on the topic of simultaneous identification of predictors and outliers. Peixoto and LaMotte (1989) introduced dummy variables for each case, which is used to detect outliers while, at the same time, variable selection is performed. Van Vuuren (2000) used the same approach and proposed a stopping rule based on the smallest true significance level (p -value) among the possible models. Hoeting, Raftery and Madigan (1996) suggested a method for simultaneous variable selection and outlier identification based on the computation of posterior model probabilities.

The aim of this paper is to investigate the feasibility of data elimination and discarding of explanatory variables simultaneously through the use of mathematical programming techniques. We believe that the combination of the two-step activity into one activity will contribute to the robustness of linear models in the sense of producing “better” linear models and thereby also enhancing the forecasting and/or explanatory capabilities of such models.

The remainder of this paper is organized as follows. In section 2 a brief background to the problem, supported by a simple example, is discussed while in section 3 the proposed mathematical programming model is presented. Section 4 contains examples from empirical data sets and section 5 outlines the bootstrap approach used to verify the prediction accuracy of the procedure generating the obtained models. Conclusions are given in section 6.

2. Background to the problem

As mentioned in Section 1, most of the approaches used to simplify a linear model involve a sequential process whereby predictors are first identified and then any possible outliers. This procedure may be criticized since the data (and possible outliers) is used to determine the choice of predictors. This may lead, in the presence of faulty data points, to a model that is unreliable. To use such a model to identify outliers can thus jeopardize the modelling process. One way of trying to overcome this situation is to investigate the discarding of outliers and choice of predictors simultaneously. Reasons for this are, firstly, the problem of masking where multiple outliers in a data set may conceal the presence of additional outliers. Secondly, one-at-a-time case diagnostics may not be suitable for situations where cases are jointly but not individually outlying and thirdly, predictors that may appear to be candidates for inclusion because of outliers. An example of how sequential procedures may produce suboptimal results can be found in Peixoto and LaMotte (1989).

To illustrate the above reservations further, consider the following function that we assume to be the underlying model:

$$y = 10 + x_1 + 3x_2 + \text{error}.$$

Table 1 contains eleven data points where this function is evaluated at fixed values for the variables x_1 and x_2 . An error was added to contaminate the data slightly. For illustrative purposes an additional variable, x_3 , was included as a possible third predictor variable.

Table 1. Data for illustrative example

No.	x_1	x_2	x_3	y	Error
1	5	7	4	36	0
2	5	10	5	47	+2
3	6	6	5	33	-1
4	8	5	4	31	-2
5	9	7	5	41	+1
6	8	9	5	45	0
7	9	5	4	36	+2
8	7	10	5	46	-1
9	6	8	2	18	-22
10	9	9	2	20	-26
11	30	40	5	160	0

It is known, from the given function, that the predictors to be selected should be x_1 and x_2 . Through inspection it is also easy to see that data points 9 and 10 (largest error) may be considered for elimination. Applying our proposed model (detailed in section 3), one can see from the results in table 2 that if only one data point is deleted together with deletion of one predictor variable, data point 11 is deleted and x_1 is discarded with R^2 -adjusted = 0.88. If, however, two data points and one predictor are deleted simultaneously, data points 9 and 10 together with x_3 are discarded as expected. This model yields an R^2 -adjusted of 0.998.

Table 2. Results of illustrative example

No. of data points deleted: 1		No. of data points deleted: 2	
Data point deleted: 11		Data points deleted: 9, 10	
Predictor discarded: x_1		Predictor discarded: x_3	
$R^2 = 0.902$	R^2 -adjusted=0.88	$R^2=0.9988$	R^2 -adjusted=0.9985
x_1 Coefficient: -	t -value: -	x_1 Coefficient: 0.964	t -value: 4.146
x_2 Coefficient: 1.392	t -value: 2.243	x_2 Coefficient: 3.024	t -value: 18.482
x_3 Coefficient: 7.763	t -value: 7.894	x_3 Coefficient: -	t -value: -

This simple example shows that it is indeed possible to produce suboptimal results if data points and predictors are not investigated (and discarded if

necessary) simultaneously. In the next section the mathematical programming model to do this is presented.

3. Mathematical programming model

In simple linear regression the screening of data may be attempted graphically but this becomes infeasible with a large number of predictor variables. It would therefore be desirable to have an automatic method of screening the data for errors and/or outliers. This can be accomplished, as will be shown in this section, by solving a mixed integer linear program by using, for example, the well-known branch and bound method and the minimum absolute deviation method (L_1 -norm) to estimate the coefficients for the predictor variables. The method could in principle also be applied to the normal least sum of squares approach, but this is less attractive since calculations become tedious and statisticians lately tend to prefer the L_1 -norm. A mixed integer linear program refers to a model in which some of the variables are restricted to integer values while others may take on continuous values. The branch and bound method is a solution technique used in integer linear programs and is based on a structured partitioning scheme and enumeration. The process is guided by bounds that are computed and used in the solution process to prune the search space. A good exposition of the technical details of integer programming models and the branch and bound algorithm can be found in Salkin and Mathur (1989).

Consider the case where we have to fit a regression function of the form

$$Y = b_0 + b_1X_1 + \dots + b_sX_s \text{ to the } n \text{ data points } (Y_i, X_{1i}, \dots, X_{si}).$$

Let us assume that we want to eliminate p data points from the regression and that we have available an upper limit K on the maximum absolute deviation from the regression equation. One possible way of solving this problem is to find the optimal solution of a mixed integer linear program where the sum

of the absolute deviations is minimised. Rabinowitz (1968) explained how the L_1 -norm can be used to find this minimum by solving the following linear program:

$$\min P = \sum_{i=1}^n (\varepsilon_{1i} + \varepsilon_{2i})$$

subject to

$$\begin{aligned} Y_i - b_0 - b_1 X_{1i}, \dots, -b_s X_{si} + \varepsilon_{1i} - \varepsilon_{2i} &= 0 \\ b_0, b_1, \dots, b_s &\text{ unrestricted in sign and} \\ \varepsilon_{1i}, \varepsilon_{2i} &\geq 0 \quad \forall i. \end{aligned}$$

The key observation that explains this is that at least one of the variables ε_{1i} and ε_{2i} will be zero in the optimal solution and $|\varepsilon_{1i} - \varepsilon_{2i}| = \varepsilon_{1i} + \varepsilon_{2i}$ since they are non-negative. Extending this model to allow for the omission of data points gives the following:

$$\min P = \sum_{i=1}^n (\varepsilon_{1i} + \varepsilon_{2i})$$

subject to

$$\begin{aligned} Y_i - b_0 - b_1 X_{1i}, \dots, -b_s X_{si} + \varepsilon_{1i} - \varepsilon_{2i} + 2K\gamma_i &= K\delta_i, \quad i = 1, \dots, n \\ \sum_{i=1}^n \delta_i &= p \\ \gamma_i &\leq \delta_i, \quad i = 1, 2, \dots, n \\ \gamma_i, \varepsilon_{1i}, \varepsilon_{2i} &\geq 0, \quad i = 1, 2, \dots, n \\ \text{and } \delta_i &\in \{0, 1\}, \quad i = 1, 2, \dots, n. \end{aligned}$$

In the solution to this problem, the p data points will be eliminated that result in the best fit of the remaining data points in the sense of minimum total absolute deviation. It is relatively easy to see that a value of K , at least as large as the optimal P (with $p = 0$), will suffice, keeping in mind that the optimal objective P is non-increasing in p . In the examples presented in section 4, K was chosen to be slightly larger than these bounds.

Consider the same case and assume that we want to eliminate r predictor variables from the equation and that we have available an upper limit L on the maximum absolute regression coefficient in any regression on a subset.

That is $L > \max(|b_1|, \dots, |b_{s-r}|)$ where the maximum is also taken over all possible subsets of $s - r$ predictor variables. The computation of L is possible but can become long and tedious and for the purpose of this study L was computed as follows. The data was scaled using a linear transformation and an ordinary regression analysis was then performed on the data set. The maximum coefficient b_i was multiplied with a large factor (arbitrarily chosen) to ensure that L complies with the stated requirement. To ensure that L was chosen large enough, the output of the model was inspected and if the model determined a coefficient equal to the chosen L , which means that L was not large enough, L was then increased and the model was solved again.

Extending the original model again to allow for the elimination of predictor variables, we formulate and solve the following mixed integer linear program:

$$\min P = \sum_{i=1}^n (\varepsilon_{1i} + \varepsilon_{2i})$$

subject to

$$\begin{aligned} Y_i - b_0 - b_1 X_{1i}, \dots, -b_s X_{si} + \varepsilon_{1i} - \varepsilon_{2i} &= 0, \quad i = 1, \dots, n \\ -L(1 - \delta_i) &\leq b_i \leq L(1 - \delta_i) \\ \delta_1 + \delta_2 + \dots + \delta_s &= r \\ \varepsilon_{1i}, \varepsilon_{2i} &\geq 0, \quad i = 1, \dots, n \\ \delta_i &\in \{0, 1\} \quad \text{for } i = 1, 2, \dots, s. \end{aligned}$$

In the solution to this problem, the $s - r$ remaining regression variables will be selected that give the best fit of the regression equation in the sense of minimal total absolute deviation.

Combining the above two models will then provide a mixed integer linear program that will eliminate data and discard predictors at the same time. This

final model, of which examples will be presented in section 4, is as follows:

$$\min P = \sum_{i=1}^n (\varepsilon_{1i} + \varepsilon_{2i})$$

subject to

$$Y_i - b_0 - b_1 X_{1i}, \dots, -b_s X_{si} + \varepsilon_{1i} - \varepsilon_{2i} + 2K\gamma_i - K\psi_i = 0, \quad i = 1, \dots, n$$

$$-L(1 - \delta_i) \leq b_i \leq L(1 - \delta_i) \quad i = 1, \dots, s$$

$$\sum_{i=1}^s \delta_i = r$$

$$\sum_{i=1}^n \psi_i = p$$

$$\gamma_i - \psi_i \leq 0$$

$$\delta_i \in \{0, 1\}, \quad i = 1, \dots, s$$

$$\psi_i \in \{0, 1\}, \quad i = 1, \dots, n$$

$$\varepsilon_{1i}, \varepsilon_{2i}, \gamma_i \geq 0, \quad i = 1, \dots, n.$$

The determination of values for p and r using the above model is a crucial aspect. Up to now, we have used the results from the model for different values of p and r to select the “best” combination that satisfies goodness of fit criteria. This amounts to the calculation of a two-dimensional grid of values as illustrated in section 4.

When dealing with small data sets (generally < 50 cases and 4 – 6 explanatory variables), the final model can easily be solved with Excel’s solver function. For larger data sets, specialized optimization software such as CPLEX (ILOG, 2002) or IBM’s OSL (OSL, 1995) should be used. All the examples discussed in section 4 were solved using the standard optimization software package CPLEX (ILOG, 2002) that can handle large numbers of variables in an integer programming model. Relatively small data sets were used in the work performed for the purpose of this paper as the objective was to study the feasibility of the suggested framework but it is accepted that the

use of larger data sets in the model may be limited by computing resources – this aspect is being investigated as part of a larger research project.

4. Examples

Three examples will be discussed briefly to illustrate the use of the proposed model. The first example is based on a data set that was used by Van Vuuren (2000) to illustrate the use of dummy variables in simultaneous variable selection and outlier detection. In the second example a data set obtained from a regression study by Roux (1994) relating GNP to various factors for different countries is investigated. The third example was chosen from Hoeting et al. (1996) with the purpose of comparing the results of the suggested mathematical programming model with their results. The three data sets are available from the website www.puk.ac.za/research/regression/index.html

A. Example 1

Van Vuuren (2000) took a data set from the literature consisting of 20 observations. The dependent variable was oxygen uptake from five chemical measurements (the predictor variables). According to the literature source the best subset model was the one based on X_3 and X_5 while observations 1, 7, 15, 17 and 20 were identified (sequentially) as influential. This model yields an R^2 value of 0.9357 and R^2 -adjusted = 0.9250. Van Vuuren's model (based on the use of dummy variables to identify predictors and outliers simultaneously) suggests that only X_3 should be included as a predictor variable, while observations 4 and 6 jointly with observations 1 and 20 were identified as influential. Table 3 presents the results obtained by using the mathematical programming model above.

Table 3. Results (Example 1: $K = 15000$, $L = 10000$)

Number of predictors discarded	Three data points deleted	Four data points deleted	Five data points deleted
1	$R^2 = 0.917$ R^2 -adjusted = 0.8895 Observations discarded: 1, 15, 19 Predictor discarded: X_1	$R^2 = 0.941$ R^2 -adjusted = 0.9125 Observations discarded: 1, 4, 6, 20 Predictor discarded: X_5	$R^2 = 0.949$ R^2 -adjusted = 0.9295 Observations discarded: 1, 4, 6, 9, 20 Predictor discarded: X_5
2	$R^2 = 0.920$ R^2 -adjusted = 0.9018 Observations discarded: 1, 7, 20 Predictors discarded: X_1, X_2	$R^2 = 0.937$ R^2 -adjusted = 0.9220 Observations discarded: 1, 7, 15, 20 Predictors discarded: X_1, X_2	$R^2 = 0.941$ R^2 -adjusted = 0.9249 Observations discarded: 1, 3, 7, 15, 20 Predictors discarded: X_1, X_2
3	$R^2 = 0.918$ R^2 -adjusted = 0.9069 Observations discarded: 1, 7, 20 Predictors discarded: X_1, X_2, X_4	$R^2 = 0.935$ R^2 -adjusted = 0.9257 Observations discarded: 1, 7, 15, 20 Predictors discarded: X_1, X_2, X_4	$R^2 = 0.938$ R^2 -adjusted = 0.9286 Observations discarded: 1, 3, 7, 15, 20 Predictors discarded: X_1, X_2, X_4
4	$R^2 = 0.879$ R^2 -adjusted = 0.8710 Observations discarded: 1, 4, 20 Predictors discarded: X_1, X_2, X_4, X_5	$R^2 = 0.919$ R^2 -adjusted = 0.9139 Observations discarded: 1, 4, 6, 20 Predictors discarded: X_1, X_2, X_4, X_5	$R^2 = 0.940$ R^2 -adjusted = 0.9361 Observations discarded: 1, 4, 6, 9, 20 Predictor discarded: X_1, X_2, X_4, X_5

From the results it can be seen that if the same number of data points (5) and predictors (3) are excluded as suggested in the literature source, the model deletes data points 1, 3, 7, 15 and 20 together with predictors X_1 , X_2 and X_4 with an R^2 -adjusted of 0.9286. It appears that when looking in a group context at the data points and predictors, data point 3 instead of data point 17 (as suggested) should be deleted.

If the model deletes four data points and four predictor variables, the same data points and predictors as in Van Vuuren's results are identified. It is, however, interesting to note that if three predictors and 4 data point are discarded observations 1, 7, 15 and 20 are discarded, while in the case of discarding four predictors and four data points, observations 1, 4, 6 and 20 are eliminated. If five data points are specified for deletion, observation 9 is also included and a better R^2 -adjusted value of 0.9361 is obtained.

B. Example 2

Roux (1994) performed a regression study relating GNP to 10 factors for different countries. This data set, consisting of 43 observations and 10 predictor variables, was used to experiment with the suggested mathematical model in the next example. Table 4 summarises part of the results obtained.

It can be noted how the predictors being discarded changed with the number of data points being deleted. For example, when we specify two predictors to be discarded together with 1 observation, X_6 and X_9 were selected. For two observations, X_6 and X_8 were discarded and for three observations, X_9 and X_{10} were selected as candidates for deletion. This finding reinforces the thought that data points and predictors have an influence on the model collectively.

Table 4. Results (Example 2: $K = 60000$, $L = 20000$)

Number of predictors discarded	One data point deleted	Two data points deleted	Three data points deleted
2	$R^2 = 0.765$ $R^2\text{-adjusted} = 0.709$ Observation discarded: 43 Predictors discarded: X_6, X_9	$R^2 = 0.794$ $R^2\text{-adjusted} = 0.743$ Observations discarded: 29, 43 Predictors discarded: X_6, X_8	$R^2 = 0.825$ $R^2\text{-adjusted} = 0.780$ Observations discarded: 1, 29, 43 Predictors discarded: X_9, X_{10}
3	$R^2 = 0.751$ $R^2\text{-adjusted} = 0.700$ Observation discarded: 43 Predictors discarded: X_3, X_5, X_{10}	$R^2 = 0.793$ $R^2\text{-adjusted} = 0.749$ Observations discarded: 29, 43 Predictors discarded: X_6, X_8, X_9	$R^2 = 0.825$ $R^2\text{-adjusted} = 0.787$ Observations discarded: 1, 29, 43 Predictors discarded: X_8, X_9, X_{10}
4	$R^2 = 0.761$ $R^2\text{-adjusted} = 0.720$ Observation discarded: 43 Predictors discarded: X_3, X_6, X_9, X_{10}	$R^2 = 0.793$ $R^2\text{-adjusted} = 0.756$ Observations discarded: 29, 43 Predictors discarded: X_6, X_8, X_9, X_{10}	$R^2 = 0.823$ $R^2\text{-adjusted} = 0.791$ Observations discarded: 1, 29, 43 Predictors discarded: X_6, X_8, X_9, X_{10}

C. Example 3

One of the examples used by Hoeting et al. (1996) was the stack loss data, taken from Brownlee (1965), and is described by them as follows. The stack loss data consists of 21 data cases that describe a chemical reaction using three explanatory variables – X_1 , a measured rate of operation, X_2 , a temperature measurement and X_3 , an acid concentration.

Hoeting et al. (1996) also quoted other authors that have already considered the stack loss data set. According to them, the general consensus is “.... that predictor X_3 (acid concentration) should be dropped from the model and that observations 1, 3, 4, and 21 are outliers”. Single deletion diagnostics for all 21 observations for the model with predictors X_1 , X_2 and X_3 provide little evidence for the presence of outliers, but robust analyses typically identify these masked outliers. The results from the method employed by Hoeting et al. (1996) concur with the general consensus and suggested that predictor variable X_3 be discarded as well as data points 1, 3, 4 and 21.

It was pointed out by them that there is some evidence that the inclusion of a quadratic term or an interaction term would lead to a better fitting model. They also quoted other authors who have suggested that transformation of the response may be appropriate. For the purpose of this paper, we have chosen, as Hoeting et al. (1996) did, not to explore these issues further.

Table 5 presents part of the results obtained from our proposed mathematical model.

Table 5. Results (Example 3: $K = 480$, $L = 20000$)

Number of predictors discarded	One data point deleted	Two data points deleted	Three data points deleted	Four data points deleted	Five data points deleted
1	R^2 -adjusted = 0.94 Observation discarded: 21 Predictor discarded: X_3	R^2 -adjusted = 0.962 Observations discarded: 4, 21 Predictor discarded: X_3	R^2 -adjusted = 0.964 Observations discarded: 3, 4, 21 Predictor discarded: X_3	R^2 -adjusted = 0.968 Observations discarded: 1, 3, 4, 21 Predictor discarded: X_3	R^2 -adjusted = 0.932 Observations discarded: 1, 2, 3, 4, 21 Predictor discarded: X_3
2	R^2 -adjusted = 0.922 Observation discarded: 21 Predictors discarded: X_2, X_3	R^2 -adjusted = 0.955 Observations discarded: 4, 21 Predictors discarded: X_2, X_3	R^2 -adjusted = 0.953 Observations discarded: 3, 4, 21 Predictors discarded: X_2, X_3	R^2 -adjusted = 0.946 Observations discarded: 1, 3, 4, 21 Predictors discarded: X_2, X_3	R^2 -adjusted = 0.964 Observations discarded: 1, 3, 4, 13, 21 Predictors discarded: X_2, X_3

Table 5 shows that when the mathematical programming model is used to discard one predictor variable and four data points, exactly the same results as with the Hoeting et al. (1996) model are obtained. Based on the R^2 -adjusted value of 0.968 this is also the “best” model.

The three examples reported in this section were chosen to show that in these cases the proposed approach is feasible to use when evaluating the omission of data points and discarding of explanatory variables simultaneously. Some methods, for example the method proposed by Hoeting et al. (1996), claim to select predictor variables and outliers simultaneously, but a two-step approach is necessary where they pre-screen the data to identify “possible outliers” which are then used in their simultaneous process. In contrast to this, the approach suggested in this paper does not need any pre-knowledge of the data; it uses standard techniques such as linear programming models, which can be implemented using standard software. Some methods usually require special software, which is privately developed and not readily available.

5. Prediction accuracy

One of the primary purposes of data analysis is to make forecasts. Once the suggested mathematical programming model has estimated the parameters of the linear model, the predictive performance has to be assessed. In other words, how accurate are forecasts made by the selected model?

A common measure of the fit of a linear model to a sample would be the sum of squared residuals

$$SSR(Model) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n \left(y_i - \sum_{j=1}^k \hat{\beta}_j x_{ij} \right)^2$$

Dividing the SSR by n gives a mean squared residual (MSR)

$$MSR(Model) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

which can be interpreted as an average of n squared errors of prediction.

The above MSR assesses prediction accuracy only for the cases in the data set that were used in fitting the prediction model. The parameter coefficients were estimated in such a way that the expected errors are minimised. This means that whenever the model is used for prediction of other (future) responses, the MSR will provide an overly optimistic estimate of the accuracy. The idea therefore is to correct the MSR or apparent error rate for this optimism. Lunneborg (2000) describes a bootstrap approach whereby an estimate is obtained for the optimism of a series of models fit to bootstrap samples. From this an aggregate optimism estimate is obtained by averaging and then used to correct the MSR. Lunneborg (2000) describes the general procedure as follows:

1. From the multivariate distribution for the random sample, $x = (X, y)$, compute the MSR for the k -parameter linear model. Note that X denotes the design matrix part of the sample and y denotes the vector of response scores.
2. For $b = 1, \dots, B$
 - (a) From the estimated population distribution, $\hat{X}$, draw a random bootstrap sample, $x_b^* = (X_b^*, y_b^*)$ of size n .
 - (b) Fit the k -parameter linear model to x_b^* , obtaining the parameter estimates $\hat{\beta}_b^*$.
 - (c) Use $\hat{\beta}_b^*$ and X_b^* to obtain response score predictions $\hat{y}_b^*$.
 - (d) Compute $a_b^* = (1/n) \sum_{i=1}^n (y_{bi}^* - \hat{y}_{bi}^*)^2$.
 - (e) Use $\hat{\beta}_b^*$ and X , the real world design matrix, to obtain response score predictions, $\hat{y}_b$.
 - (f) Compute $t_b^* = (1/n) \sum_{i=1}^n (y_i - \hat{y}_{bi})^2$.
 - (g) Compute and save $o_b^* = (t_b^* - a_b^*)$.

3. Compute $\hat{O} = (1/B) \sum_{b=1}^B o_b^*$, the estimated optimism of the MSR.
4. Compute the estimated true prediction error rate, $TPE = MSR + \hat{O}$.

The mathematical model suggested in this paper is based on the minimum absolute deviation and to illustrate the true prediction error rate methodology all the squared calculations, reflecting the normal least sum of squares approach, were changed to absolute deviation formulas and the mean absolute deviation (MAD) was corrected (see step 4 of the procedure above).

We illustrate the optimism correction with the data from example 3 using the (best) model where one predictor variable and four data points were discarded. The obvious changes to the bootstrap formulas were made to reflect the data points deleted. The MAD for this model was calculated as 0.87. We can interpret this as the amount by which, on average, Y_i would be error prone when predicting it from our model. Setting $B = 50$ in the bootstrap algorithm a bootstrap estimate of the optimism of MAD was calculated as 0.68. After increasing the original MAD by this amount, we can expect to mispredict Y for new cases by about 1.55. The MAD for the original data set (using an L_1 regression model and including all predictor variables and data points) is 2.18. This confirms that higher prediction accuracy for the mathematical programming model exists as expected when data points and predictor variables are discarded.

It can be argued that the more data points deleted, the better the model would be. From the graph in figure 1 it can be seen that the MAD indeed decreases, but taking the true prediction error rate into account a lowest turning point is reached where after the error will start increasing. This is mainly due to the fact that the optimism term $\hat{O}$ grows larger if more data is eliminated. The turning point occurs at the model where one predictor variable and four data points are discarded.

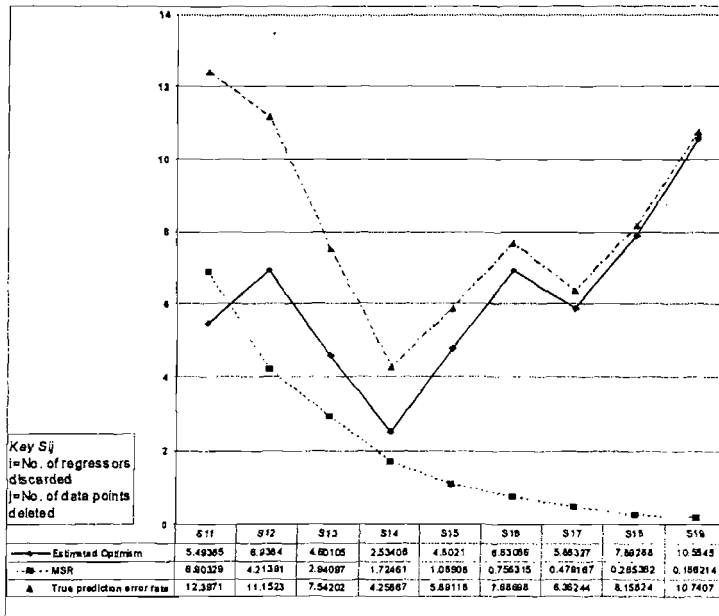


Figure 1. True prediction error using MAD

For completeness sake the optimism correction was also computed using the least squares approach. Figure 2 shows a graphical representation of the results. In this case the MSR was calculated as 1.72 and the optimism correction as 2.53, which suggested that we could expect to misinterpret Y for new cases, in squared error sense, by about 4.25. It is significant that the turning point again occurs where four data points are discarded.

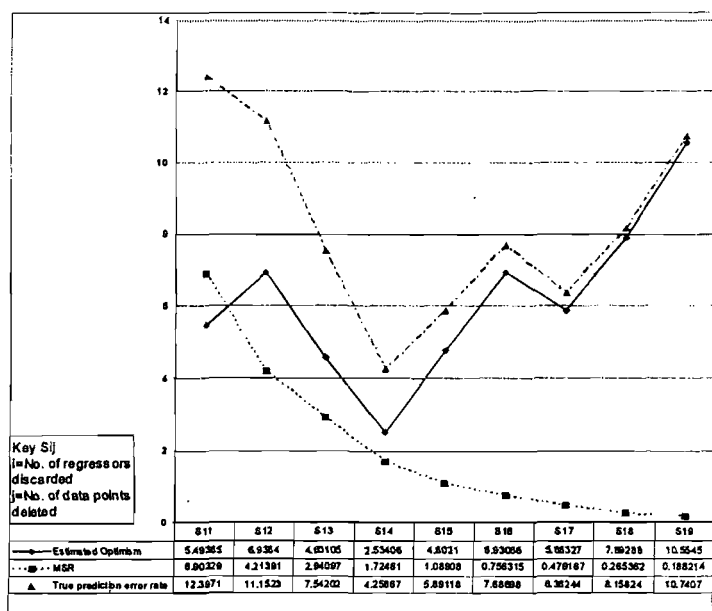


Figure 2. True prediction error using MSR

A limited and basic simulation was performed to support the idea of computing the prediction accuracy using the bootstrap method described above. It was decided not to generate artificial data sets with contaminated data points to which the model can be applied but rather generate data sets from an existing data set. The stack loss data set from example 3 was therefore used to generate 10 different data sets each containing 15 randomly selected cases. A bootstrap estimate of the prediction accuracy for each of the 10 data sets was computed and the results are compared with the MAD in table 6.

Table 6. Simulation results

Simulation	MAD	K-value in model	L-value in model	Lowest true pre- diction error rate	Lowest true prediction error rate occurred when discarding the following
1	1.73	347	20000	1.67	1 regressor and 3 data points
2	1.17	455	20000	0.74	1 regressor and 3 data points
3	1.61	382	20000	1.34	1 regressor and 2 data points
4	1.83	386	20000	1.72	1 regressor and 5 data points
5	1.67	367	20000	1.69	1 regressor and 3 data points
6	1.81	383	20000	1.85	1 regressor and 2 data points
7	1.41	389	20000	0.69	1 regressor and 2 data points
8	1.57	383	20000	1.22	1 regressor and 2 data points
9	1.73	421	20000	1.61	1 regressor and 2 data points
10	1.76	387	20000	1.38	1 regressor and 3 data points

It should be noted that the lowest true prediction error rate, in the 2nd last column of table 6, does not always occur at the same number of data points deleted – this is due to the sample selection process in the simulation. In general the true prediction error rate of the new model is better than the MAD of the original model which supports the idea of discarding data points and explanatory variables simultaneously using the proposed model.

6. Conclusion

A mathematical programming technique was proposed to assist with the development of robust linear models by discarding data and predictor variables simultaneously.

Results such as in tables 3, 4 and 5, as well as experience with a few other data sets (that we do not report on here) suggest that it is possible and desirable to investigate data elimination and discarding of explanatory variables simultaneously. The results of the mixed integer linear programming model compare favourably with other analyses performed on other data sets, and the predictive accuracy of the selected models is improved as shown by the resampling technique in section 5.

Future work to be done is an investigation into the possibility of designing an algorithm that can handle large data sets. The branch and bound method requires a lot of computing resources and one possibility might be to apply the proposed model to subsets of large data sets. The algorithm will also have to provide for determining a way to decide at least an approximation of the number of predictor variables and data points to be discarded in order to reduce the computational burden of trying many combinations. In addition, the use of R^2 -adjusted as criterion to determine the “best” combination of data points and predictor variables to be discarded, should be investigated further.

Evidence was found where R^2 -adjusted proved to be too insensitive (especially in larger data sets) to be used as an indicator of how many data points are to be deleted.

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